California State University Long Beach

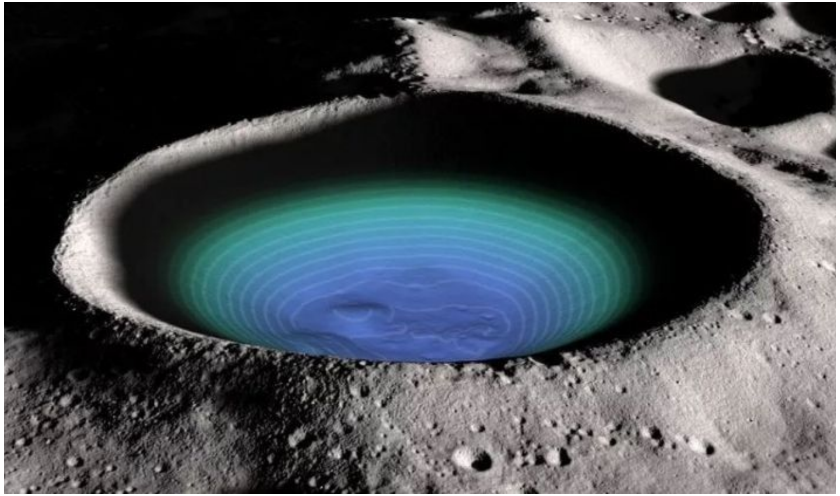
Team Auger

Project Proposal

September 04, 2023

Agenda

- Team Introductions
- Proposed Project Overview
  - Project Mission
  - Mission Profile
  - Proposed Level I Requirements
  - Proposed Design Approach
  - Subsystem Overview
- Logistics
  - Rough Project Timeline
  - Proposed Deliverables to Customer
- Conclusion
  - Proposed Project Justification



https://www.indiatimes.com/technology/science-and-future/nasa-moon-water-discovery-526143.html

Team Introduction

Alex Contreras
Team Lead & Heat Transfer Lead

Experience in Power Budgets, Subsystem Integration, Design Reviews

Zachary Tep
Avionics Lead

Experience in structure, manufacturing and propulsion.

Alondra Barba
Structures Lead

Experience in structure, design, and manufacturing

David Guerra
GNC Lead

Experience in structures, fluids mechanics and aerodynamics.

Erick Nguyen
Mechanical Design lead

Experience in mechanical design and structural analysis

Bryan Ramos
Telecommuni-cations lead

Experienced in structural analysis, fluids and Stability/control

Damon Nguyen
Power Lead

Experience in thermodynamic, structures, aerodynamics, and fluids.

Project Overview

Problem:

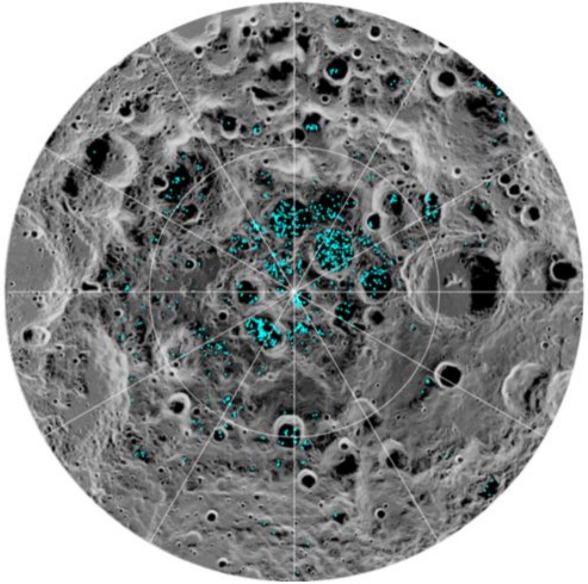
- The lack of reliable and efficient methods to extract water ice from the lunar surface poses a significant obstacle for establishing a sustainable human presence on the Moon.

Challenge:

- Water ice is found to be most abundant at the Moon's poles, extremely cold temperatures, and rough terrains makes it difficult to extract water ice from the lunar surface.

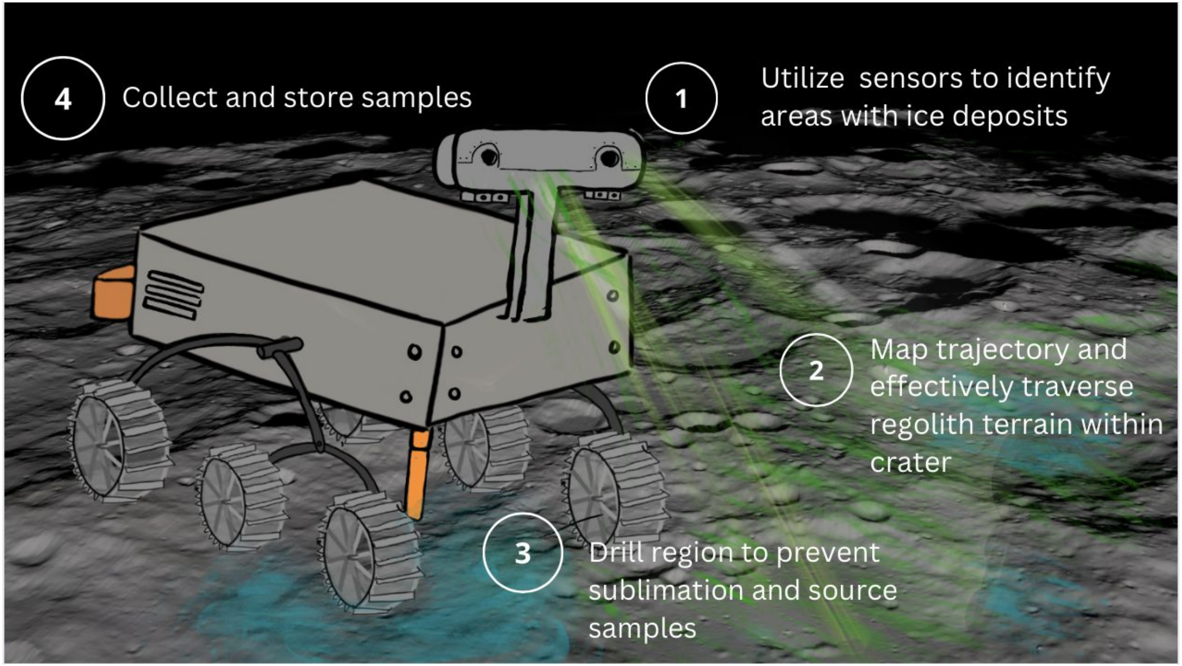
Solution:

- Design an autonomous lunar rover that demonstrate its ability to extract water ice on the lunar surface.



The distribution of surface ice (marked in blue) at the Moon's south poles. For more information visit.

Mission Profile



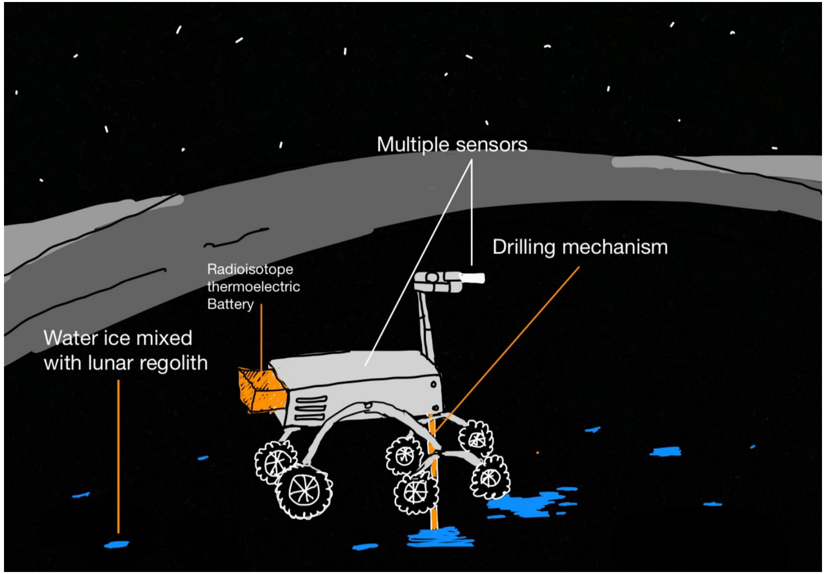
Level I Requirements

- **Requirement 1:** The rover shall autonomously identify areas in the permanently shadowed region with water ice deposits with an 80%±5 accuracy (TBD).
- **Requirement 2:** The rover shall collect a minimum of 20 samples of 100g (TBD) of water ice over the entire mission life cycle.
- **Requirement 3:** The rover shall maintain the collected water at a temperature of 75℉±5 (TBD) in a pressurized storage tank to avoid sublimation.
- **Requirement 4:** The rover shall be operational for a minimum of 50 days (TBD).
- **Requirement 5:** The rover shall travel a minimum distance of 10km (TBD).

Proposed Design Approach

Design Approach:

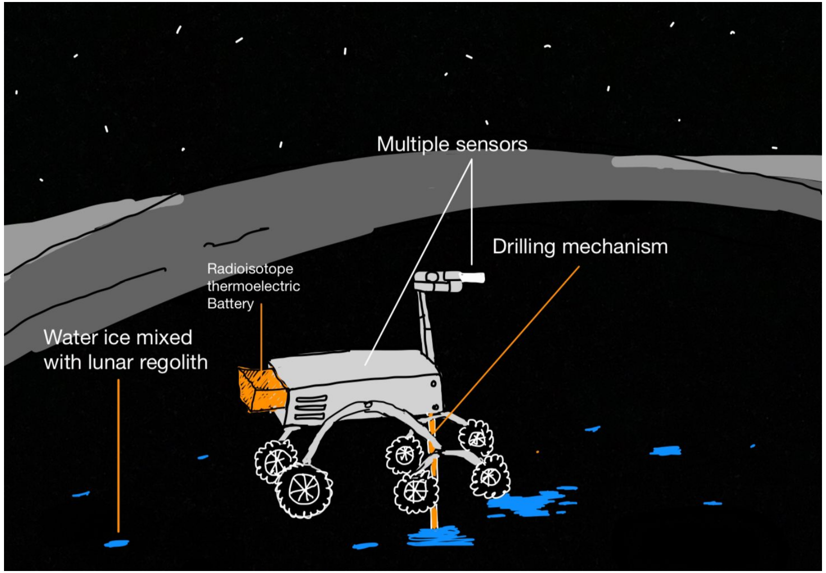
The autonomous lunar rover will be powered by a Radioisotope Thermoelectric Generator (TBR). It will feature a six-wheel system operating independently from each other, multiple sensors/camera, a pressurized tank storage system, drilling capability and a communication system.



Proposed Design Approach

Design Approach (Continued):

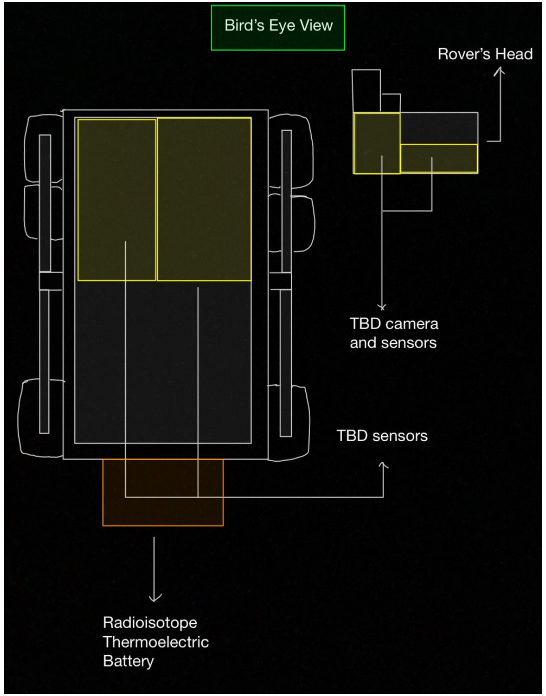
- Build a System Acceptance Test (Flat SAT)
- Build and test Engineering Model (EM)
- Qualification Unit
- Flight Ready Unit



Subsystem: Avionics

Avionics Subsystem Goals:

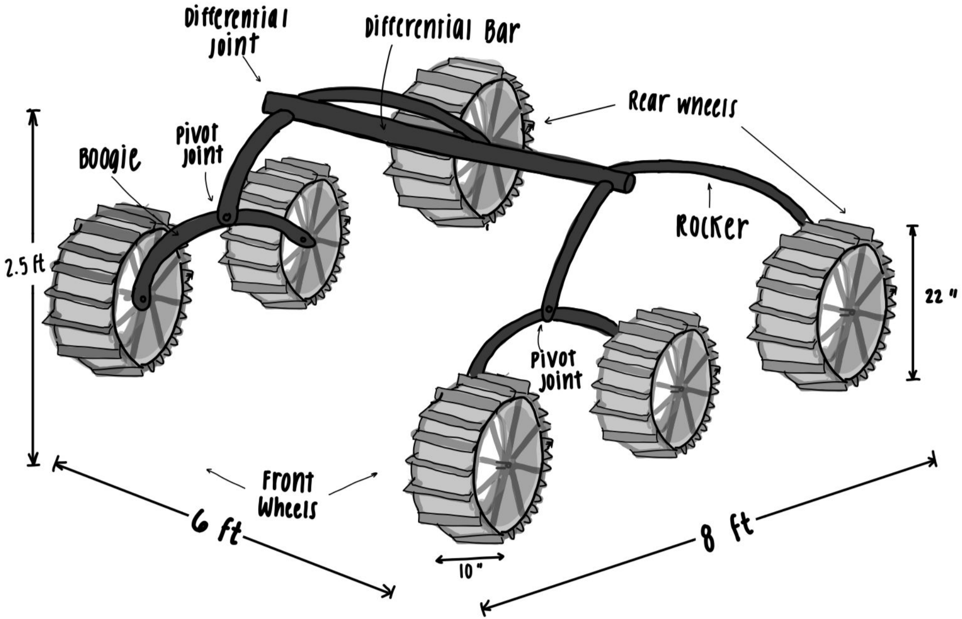
- Seamlessly integrate data from multiple sensors
- Scan multiple areas within the crater where water ice is most likely to be abundant
- Utilize TBD sensor to differentiate water ice from lunar regolith samples
- Coordinate effectors to execute precise movement tasks, such as drilling
- Utilize a TBD sensor to measure the amount of water ice collected



Subsystem: Structures

Structure Subsystem Goals:

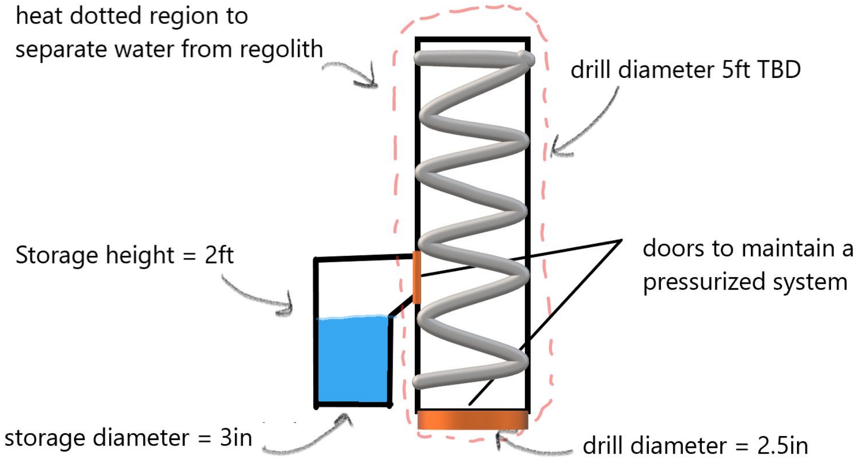
- Use TBD material to withstand freezing temperatures as low as -200℃ (TBD)
- Provide weight support and attachments for each subsystem
- Utilize TBD wheel material to facilitate movement over varying regolith terrains
- Utilize TBD suspension system to provide stability



Subsystem: Mechanisms

Mechanisms Subsystem Goals:

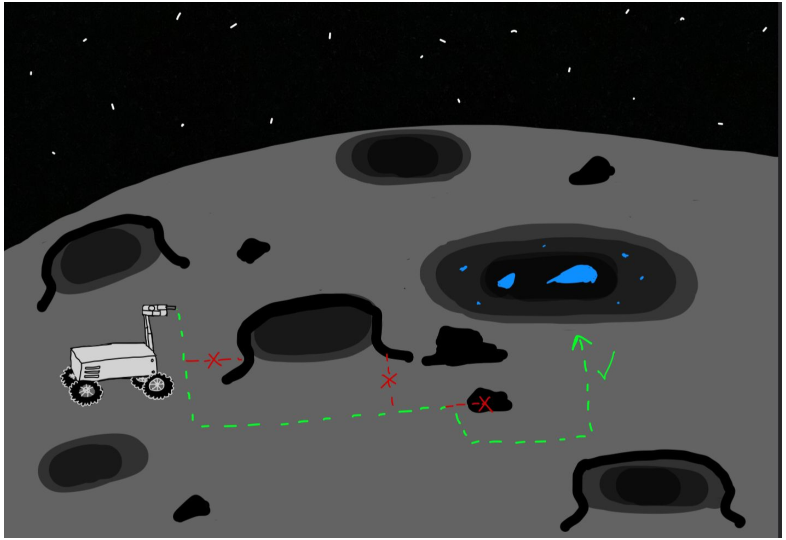
- Design a drilling mechanism that's optimized for drilling through water ice
- Store a minimum of 2000g of water ice within the rover
- Ensure sufficient pressure for water storage and drill region to prevent sublimation
- Doors ensure a seal to prevent exposure to vacuum



Subsystem: GNC

Guidance, Navigation, Control Subsystem Goals:

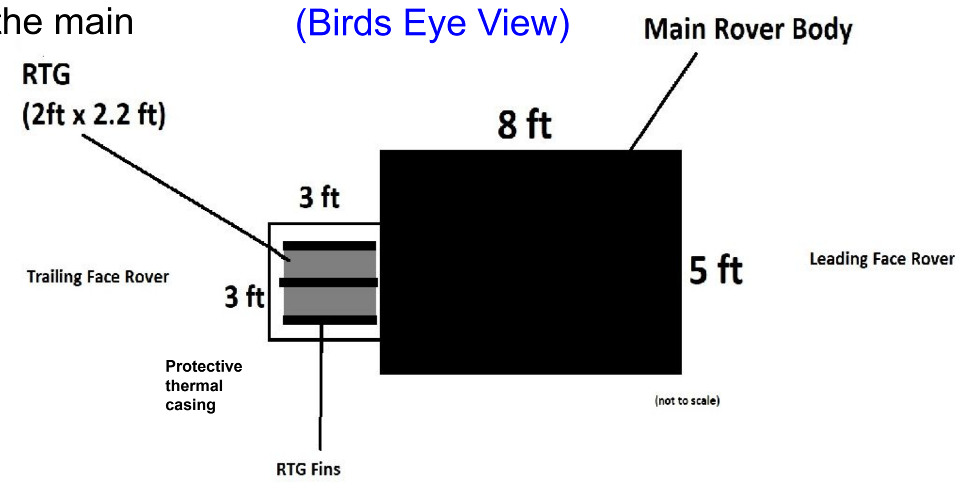
- Provide the location of the rover as it traverses through the moon's surface up to 1.5 m accuracy.
- Utilize path planning algorithm, to plan the next trajectory up to 2 meters per sequence.
- Create a map of the traveled trajectory for future use.
- Capture images of the terrain for visualization.



Subsystem: Power

Power Subsystem Goals:

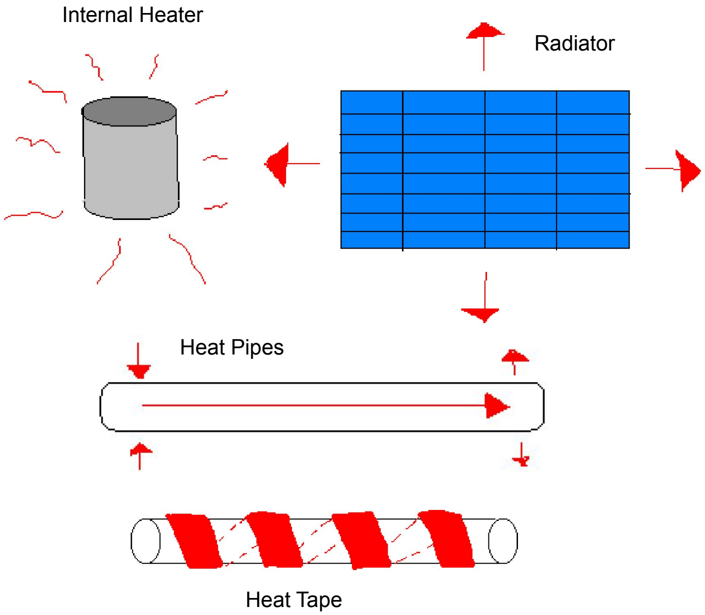
- Use Radioisotope Thermoelectric Generators as the main power source
  - converting heat from radioactive isotopes
- Provide required power to all subsystems during all operational modes
- Protect instrumentation from power fluctuations (Overvoltage and Overcurrent)



Subsystem: Heat Transfer

Heat Transfer Subsystem Goals:

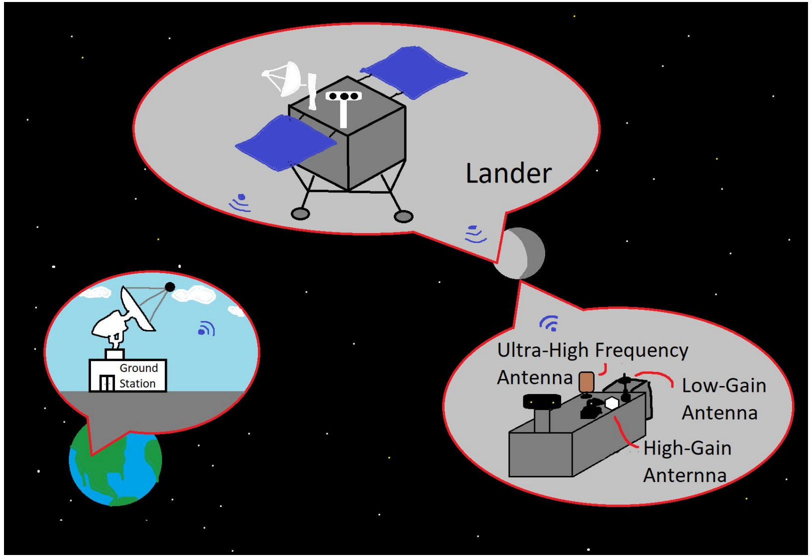
- Withstand the permanently shadowed craters' cold temperatures (both internal components and external mechanisms)
- Utilize heat controlling systems to sustain durations in the crater, prevent freezing/overheating
- To prevent surface water sublimation, guarantee that the rover's heating does not influence the surface temperature of the ground



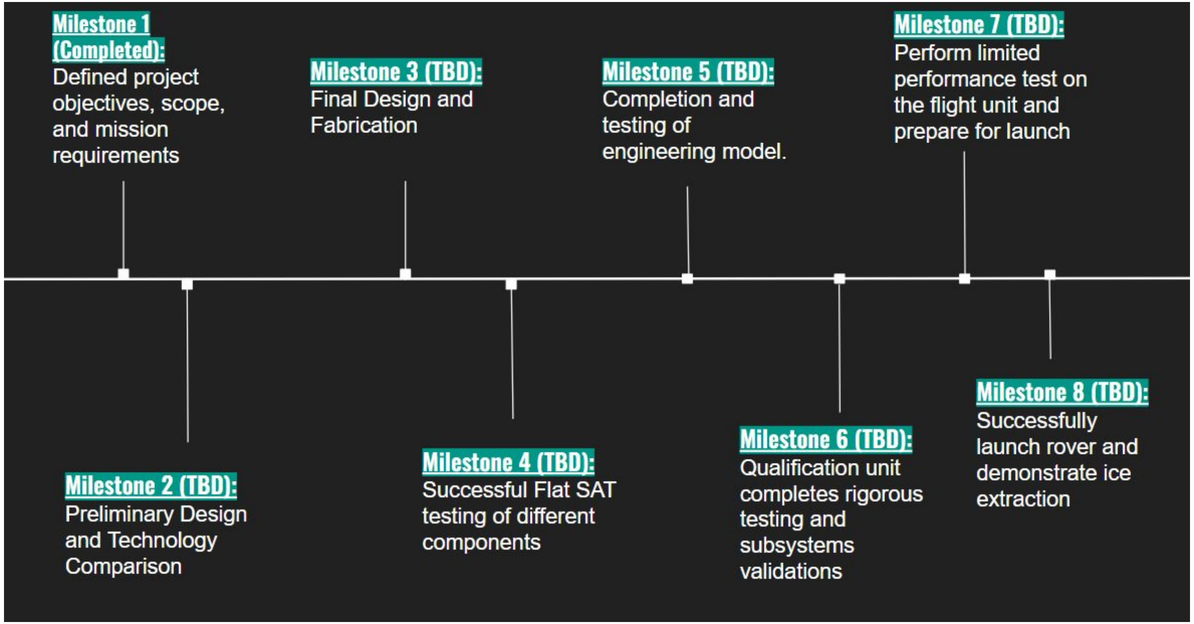
Subsystem: Telecommunications

Telecommunications Subsystem Goals:

- Land on the surface of the moon near the south pole to have direct line of sight to Earth
- Use radio waves in order to communicate between the rover and lander.
- Relay data transmissions from the lander to ground stations



Logistics


Milestone 1 (Completed):
Defined project objectives, scope, and mission requirements
Milestone 2 (TBD):
Preliminary Design and Technology Comparison
Milestone 3 (TBD):
Final Design and Fabrication
Milestone 4 (TBD):
Successful Flat SAT testing of different components
Milestone 5 (TBD):
Completion and testing of engineering model.
Milestone 6 (TBD):
Qualification unit completes rigorous testing and subsystems validations
Milestone 7 (TBD):
Perform limited performance test on the flight unit and prepare for launch
Milestone 8 (TBD):
Successfully launch rover and demonstrate ice extraction

Conclusion

- One of the most essential compound for sustainable life as humans is water. With space travel gaining more technological advancements by the day, there will eventually be a need for humans to be fully sustainable in space and potentially other planets.

- The discovery of water ice on our moon could be exactly the answer we are looking for. Through a rover, extracting this water ice would be a major advancement to sustain human exploration on the Moon and other planets in the future.

Concerns: Our missions main concern would be to withstand extreme cold conditions in the permanently shadowed area, which is located at the South Pole on the Moon to investigate the concentration of water ice.

References

"Mars 2020 Perseverance Launch Press Kit." *NASA*, NASA, www.jpl.nasa.gov/news/press_kits/mars_2020/launch/mission/spacecraft/power/. Accessed 4 Sept. 2023.

Mahoney, Erin. "Moon's South Pole Is Full of Mystery, Science, Intrigue." *NASA*, NASA, www.nasa.gov/feature/moons-south-pole-is-full-of-mystery-science-intrigue. Accessed 17 Aug. 2022.

"The Rover's Temperature Controls." *NASA*, NASA, www.mars.nasa.gov/mer/mission/rover/temperature/#thermostats-and-heat-switches. Accessed 4 Sept. 2023.

"RIMFAX."*NASA,* NASA, https://mars.nasa.gov/mars2020/spacecraft/instruments/rimfax/#:~:text=The%20Radar%20Imager%20for%20Mars'%20Subsurface%20Experiment%2C%20known%20as%20RIMFAX,the%20ground%20under%20the%20rover. Accessed Sept. 4th, 2023.

References

"Guidance, Navigation and Control Systems." *NASA,* NASA, https://mars.nasa.gov/mro/mission/spacecraft/parts/gnc/ Accessed Sept. 4th, 2023.